



State of Ohio Administrative Policy

No: MARCS-01
MARCS

Effective:
May 27, 2025

Use of MARCS Communication Towers, Shelters, and Grounds (MARCS Co-Location)

Issued By:

Kathleen C. Madden, Director

I. Purpose

The purpose of this policy is to formalize the process and requirements for entities to co-locate equipment on Multi-Agency Radio Communication System (MARCS) towers. The first occurrence of a defined term in the policy is in bold, italic type, and is hyperlinked to the definition in Section IV.

II. Scope

This policy applies to all entities, both public and private, seeking use of MARCS tower sites, including the tower, shelter, and/or grounds owned, leased, or otherwise legally controlled by MARCS.

III. Policy

The overarching priority of MARCS is to preserve future necessary expansion of MARCS equipment on the tower structure. Any tower analysis will first consider the designed-in future MARCS loading and preserve it. First priority for tower co-location will be given to state agencies utilizing MARCS as their primary wireless radio system, and then to public safety, police, fire, and emergency medical service (EMS) providers. Second priority for tower co-location will be given to commercial wireless carriers, such as cellular or broadband, and private enterprises with wireless radio needs. Commercial wireless carriers and private enterprises shall not co-locate in MARCS shelters. No entity shall co-locate any equipment on MARCS property without the written consent of MARCS and without first entering into the MARCS License Agreement.

- A. **Application Process:** Entities wishing to co-locate on a MARCS tower must complete and submit the Tower Site Co-Location Application, Attachment A to this policy. Upon receipt and review of the Application, MARCS will provide an initial approval or denial of the

Application. Initial approval will be granted if the tower has not been previously determined to be unable to support additional equipment and there is a demonstrated benefit to MARCS, a law enforcement entity, or a local community. An application to co-locate on a tower that has previously been determined to be unable to support additional equipment will be denied. The Application is attached to this Policy.

- B. **Available and Excluded Towers:** Applicants may request a list of towers with potential co-location availability from MARCS.
- C. **Tower Analysis and Intermodulation Study:** After receiving initial approval, entities will have both a **Structural Tower Analysis** and **Intermodulation Study** conducted.
 - 1. Entities shall engage an engineer to conduct a Structural Tower Analysis utilizing a **Class III Rating**. The Structural Tower Analysis shall produce an engineer-stamped letter indicating the tower's ability to hold current loads, MARCS-design future loads, and the additional requested loads, unless otherwise specified. This Structural Tower Analysis will be conducted at the entity's expense.
 - 2. MARCS may conduct an Intermodulation Study at no cost to the entity, if requested. Otherwise, the entity must conduct the Intermodulation Study at its own expense.
 - 3. Entities shall provide the Structural Tower Analysis and Intermodulation Study to MARCS for approval and acceptance. After an entity's application has been approved, copies of all studies, drawings, plan modifications, as built, and analysis shall be provided to MARCS.
- D. **Technical and Site Requirements:** A co-locating entity shall meet all technical and site co-location requirements:
 - 1. Shelter Equipment
 - a) Equipment shall be installed in 7'x19" equipment racks unless otherwise specified.
 - b) Equipment shall be grounded to the interior halo ring per Motorola R56 standard.
 - c) Equipment shall be on a dedicated air conditioning (AC) breaker using the non-uninterrupted power supply (UPS) breaker panel.
 - d) AC power connections shall be installed using the ceiling electrical raceway.
 - e) No equipment shall use the existing shelter wall outlets or connect to the MARCS UPS breaker panel.
 - f) Equipment shall be installed in a manner consistent with the installation of the MARCS equipment.
 - g) New equipment British thermal unit (BTU) loading shall be provided to MARCS to determine if the existing heating, ventilation, and air conditioning (HVAC) size is satisfactory.
 - h) New equipment power loading shall be provided to MARCS to determine if the existing generator size is satisfactory.

2. Antenna and Transmission Lines
 - a) Antennas shall be mounted using a Pirod side arm or its approved equivalent.
 - b) Installation shall be completed as directed by manufacturer's specifications.
 - c) Transmission lines shall be grounded at the transmission line port following manufacturer's specifications.
 - d) Additional grounding shall be completed at the following locations:
 - (1) On the tower's bottom ground buss bar
 - (2) Every 200' from the bottom ground buss bar
 - (3) At the antenna base
 - e) A Polyphaser coaxial protector or its approved equivalent shall be installed on each transmission line according to manufacturer's specifications.
 - f) Transmission lines shall be supported on the tower and transmission line bridge consistent with the method currently in place by MARCS.
 - g) Only one transmission line per port shall be allowed unless otherwise specified. If needed, additional ports shall be added at co-locator's expense.
3. Site Entry
 - a) A personnel list detailing potential site visitors shall be provided by the entity co-locating. This list shall include full name, date of birth, and social security number for each potential site visitor. Each potential visitor submitted on the list shall be subject to an electronic record check, and/or a ten-print fingerprint check, conducted by the Ohio State Highway Patrol. This list shall be updated annually or as changes occur.
 - b) Sites are alarmed. All personnel shall call the MARCS Network Operations Center (NOC) at 1-866-646-2727 before entering the gate and immediately before departure of the compound.
 - c) Anyone entering the site shall provide their name, company, contact information, and reason for entry.
4. Shelter Entry
 - a) Shelters are alarmed. All personnel shall call the MARCS NOC at 1-866-646-2727 before entering the gate and immediately before departure of the compound.
 - b) Anyone entering the shelter shall provide their name, company, contact information, and reason for entry.
 - c) Shelter entry keys are available by contacting the MARCS office at 614-995-0060 or emailing MARCS.admin@das.ohio.gov. Keys are issued after background checks and keys shall be reprogrammed yearly.
5. Administrative Requirements
 - a) Copies of all documentation, including a site plan for the co-location incorporating both the proposed equipment and the existing MARCS facilities, tower/site elevations, antenna details, shelter plans (including foundations and/or caissons), utilities and a grounding plan, shall be provided to MARCS by the entity.
 - b) Any work which is to be completed on a MARCS tower site must be approved by the MARCS Program Director or designee. To obtain approval, contact the MARCS NOC with the work to be completed, the name of the company or individual to complete the work, and date/time of access at least 48 hours prior

to the date the work is to be completed except for emergencies. Email: das.marcsnoc@das.ohio.gov.

- c) Any and all costs associated with co-location are the sole responsibility of the entity co-locating, unless otherwise specified in the MARCS License Agreement.
- d) No equipment shall become operational until all requirements contained herein have been met and an inspection of the site has been completed and approved by a MARCS site engineer.
- e) Equipment tampering by any entity co-locating, or their designee, shall be subject to criminal investigation and prosecution to the fullest extent of the law. Any equipment tampering shall be grounds for immediate eviction from the site.
- f) These requirements for co-location are in addition to any other terms and conditions of the MARCS License Agreement with the entity co-locating.
- g) All entities co-locating must additionally follow terms and conditions set forth in the MARCS License Agreement.
- h) Changes to initial installation to shelter/site shall be approved by MARCS technical staff prior to changes. Changes shall not be made prior to MARCS approval.
- i) Violators of this policy shall be subject to disconnection of service and removal of equipment.
- j) All equipment installed inside the MARCS shelter shall be labeled with the following information:
COMPANY NAME: _____
EMERGENCY CONTACT NAME: _____
EMERGENCY CONTACT PHONE: _____
- k) A current copy of co-locators' FCC license shall be submitted to MARCS prior to installation of any equipment.

E. Fees for Co-Location

- 1. State agencies and tier partners utilizing MARCS as their primary wireless radio system shall pay no fee for utilization of MARCS towers or shelters.
- 2. Non-State public safety agencies shall be charged a fee of \$150 per month per antenna, if utilizing the on-site MARCS shelter. Use of the MARCS shelter shall include MARCS provided commercial and back-up power.
- 3. Non-State public safety agencies shall be charged a fee of \$125 per month per antenna, if utilizing their own or other non-MARCS on-site shelter.
- 4. Commercial carriers and/or other non-public safety/public service entities shall be charged a negotiated rate. MARCS will provide an estimate of the co-location license fee at the time an entity submits its application. The negotiated rate will take into consideration the following factors:
 - a) Height of the tower;
 - b) Location of the entity on the tower;
 - c) Amount of load of the entity's equipment;

- d) Location of the tower;
- e) Whether the entity will use MARCS' shelter or power.

IV. Definitions

- A. Class III Rating. Under the American National Standards Institute and Telecommunications Industry Association, Class III is the highest reliability rating and is assigned to structures used primarily for essential communications and where a failure represents a substantial hazard to human life and/or damage to property.
- B. Intermodulation Study. An intermodulation study calculates all possible frequencies that could potentially interfere with receivers at a communications site.
- C. Structural Tower Analysis. A structural tower analysis uses engineers to determine how a tower will respond to various loads and stresses to help identify the stability of the structure.

V. Authority

2023 Am. Sub. H.B. 33, Section 213.10

VI. Resources

None.

VII. Inquiries

Direct inquiries about this policy to:

MARCS Program Office
Ohio Department of Administrative Services
4200 Surface Road
Columbus, Ohio 43228

614.995.0060 | das-MARCS.admin@das.ohio.gov

State of Ohio Administrative Policies may be found online at
<https://das.ohio.gov/home/policy-finder/filter-policy-finder>

VIII. Revision History

Date	Description of Change
05/27/2025	Combined MARCS MPP 17.0 and MARCS MPP 19.1 into new policy template. Adds definitions and removes duplication of information. Co-Location Application updated and added as an attachment to the policy.